



Filtered-error RLS for self-tuning disturbance feedforward control with application to a multi-axis vibration isolator[☆]

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ABSTRACT

High performance vibration isolation can be realized by disturbance feedforward control with a self-tuning generalized FIR filter and residual noise shaping. For this application, filtered-error recursive least squares (FeRLS) self-tuning is proposed in a multi-input multi-output context. In comparison to filtered-error least mean squares (FeLMS), FeRLS achieves faster and more uniform parameter convergence without the need of pre-whitening. Efficient implementation is realized by exploiting sparsity in the involved matrices. Feasibility of implementation is demonstrated on a multi-axis hard-mount vibration isolation setup. Experimental results show the better parameter convergence and the ability to track changes in the floor vibration spectrum. A reduction of the transmissibility of floor vibrations up to 40 dB in the frequency range of interest is obtained, reducing vibration power by 90–94% in the 1–300 Hz frequency band in multiple directions.

1. Introduction

Vibration isolation is employed in many high-precision machines to reduce the effect of floor vibrations on precision [1,2]. Active vibration isolation avoids the need for high masses typical for passive vibration isolation. A hard-mount suspension with disturbance feedforward control combines a low compliance to internal forces with a low transmissibility of floor vibrations [3]. Disturbance feedforward control (DFC) [4], also referred to as active vibration control [5], employs floor-mounted sensors to generate actuator forces that counteract the effect of floor vibrations. The required counter force is predicted from a model of the isolator dynamics.

A self-learning filter enhances model accuracy, which improves vibration isolation performance. Typically, variants of the least mean squares (LMS) algorithms are used, such as filtered-reference LMS (FxLMS, [6,7]), filtered-error LMS (FeLMS, [8,9]), and filtered-U LMS (FuLMS, [10,11]). The main advantage of these algorithms is the low computational load. However, convergence speed is typically low, which can be attributed to large eigenvalue spread in the so-called covariance matrix. Various methods have been proposed to reduce this issue; Pre-whitened LMS [8] and normalized LMS (NLMS, [12]) reduce the eigenvalue spread for colored input signals, while eigenvalue spread due to plant dynamics can be reduced with modified FxLMS [8,13] and generalized filters with pre-defined controller poles stored in orthonormal basis functions [14,15]. Residual noise shaping was proposed

previously [16] to reduce the effect of modeling errors and poor signal-to-noise ratio (SNR). Although these methods lead to improved convergence, these LMS algorithms still minimize the instantaneous squared error instead of the mean squared error (MSE), which leads to a steady-state variance on the estimated parameters. This is particularly severe in case of undermodeling or poor SNR in combination with a large adaptation rate. The variance in the parameters can be reduced by lowering the adaptation rate, but this slows down convergence.

The steady-state variance on the parameters can be reduced by minimizing the MSE, which can be obtained through recursive least-squares (RLS) algorithms. A further advantage of RLS is the fast convergence in mean without the need of pre-whitening and orthonormal basis functions [14]. Similar to LMS, filtered-reference RLS (FxRLS) and filtered-error RLS (FeRLS) can be used to deal with plant dynamics [17,18]. The use of FxRLS has been proposed for active noise control [19,20] and active vibration isolation [21]. However, FxRLS is not preferred for multi-input multi-output systems because of the use of the computationally heavy Kronecker product. Therefore, in this paper FeRLS is proposed for a multi-axis vibration isolation system. Although FeRLS no longer minimizes the original output error, but the filtered output error, it is shown previously that good vibration isolation can be realized for high-precision machines using residual noise shaping [16].

The main contribution of this paper is the application and experimental validation of FeRLS for active vibration isolation on a

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multi-axis vibration isolation system. FeRLS updates the coefficients of a generalized FIR filter and residual noise shaping is used to focus the optimization for the relevant frequency band and to negate the effect of modeling errors. The paper extends the prior proposal of FeRLS for active vibration isolation [22] to multi-axis systems and adds on the inclusion of a forgetting factor and experimental validation. An efficient implementation of FeRLS for a multi-axis system is proposed exploiting sparsity of the regression matrix. This enables real-time implementation, which is validated experimentally. The experimental setting and approach used for FeRLS in this paper are similar to the prior research using FeLMS [16], but extended by including the pole of the actuator dynamics, enabling minimization of the error in a wider frequency band. Parameter convergence, disturbance tracking and vibration isolation performance are shown and compared to the FeLMS case.

The remainder of this paper is organized as follows. A description of the experimental setup for active vibration isolation is presented in Section 2. The problem formulation is found in Section 3, whereas the self-tuning controller with FeRLS is presented in Section 4. Experimental results comparing the proposed FeRLS method with an existing FeLMS method are presented in Section 5, and the main conclusions are summarized in Section 6.

2. Hard-mount vibration isolation system

The vibration isolation strategy in this paper is intended for precision machinery such as equipment for wafer inspection in semiconductor industry. Conventionally, these systems rely on soft-mount isolators for suppression of floor vibrations [23]. In this paper we use hard-mount isolators to provide a better stiffness to counteract direct disturbance forces acting on the platform, but needing an active system for suppression of floor vibrations [23]. The vibration isolation setup that will be used throughout this paper is shown in Fig. 1(a). This setup has six suspension frequencies that are representative for a hard-mount isolator [23]. The suspension frequencies are between 15 and 45 Hz. The isolated part of the setup consists of a platform body with a mass of 5.4 kg connected to six voice coil motors (VCMs, type Geepus VM4032-250) via wire springs in the configuration resembling a Stewart platform. The VCMs are provided with the actuator voltages $u(t) \in \mathbb{R}^6$ and are guided by circular leaf springs. The VCMs are mounted via bridge structures on a rectangular floor plate. The floor plate is equipped with three accelerometers (type Endevco 7703A-1000), measuring the vertical floor plate accelerations denoted by $a_0(t) \in \mathbb{R}^3$. Moreover, six accelerometers (type Endevco 7703A-1000) are attached to the Stewart platform, measuring the platform accelerations, denoted by $a_1(t) \in \mathbb{R}^6$, in the actuator's directions. The sensors are connected to signal conditioners containing a second-order high-pass filter at 0.1 Hz and a second-order low-pass filter at 3000 Hz to filter sensor noise. The controllers are implemented on a dSPACE DS-1005 digital signal processor running at a sampling frequency of $f_s = 5000$ Hz.

To simulate an industrial environment, the floor plate can be excited using three vertical piezo shakers generating three independent random noise excitations. For the experiments in this paper, two different floor spectra have been considered, see Fig. 2. The first spectrum is referred to as “Flat” and applies excitations with an approximately flat power spectral density comparable to the frequently used BBN-B/C curves [24,25]. The second spectrum is referred to as “Colored”, and has enhanced frequency power between 40–90 Hz and around 150 Hz. The results of these experiments are further discussed in Section 5.

2.1. Feedback control

The focus of this paper is feedforward control. Nonetheless, feedback control is added for skyhook damping [26] of the suspension modes. This damping reduces the amplitude of the payload modes, which have only little mechanical damping, to make sure that the

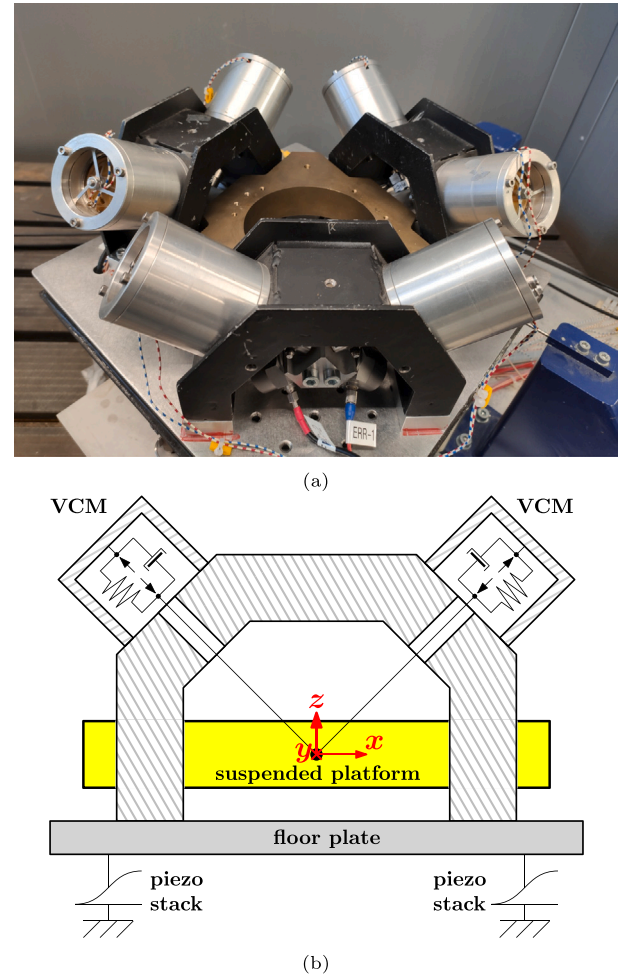


Fig. 1. (a) Photo of the test setup with the brass platform suspended to the cylindrical actuators, which are connected to the floor plate via the black bridge structures [23]; and (b) schematic picture including the coordinate frame of the elastic center; the VCMs provide actuator voltages u and are suspended in circular leaf springs providing passive stiffness and damping. The floor plate is excited using piezo actuators and equipped with six accelerometers measuring a_0 of which only the vertical components are used in this paper. The platform body is equipped with six accelerometers measuring a_1 , which are aligned with the actuator's line of action.

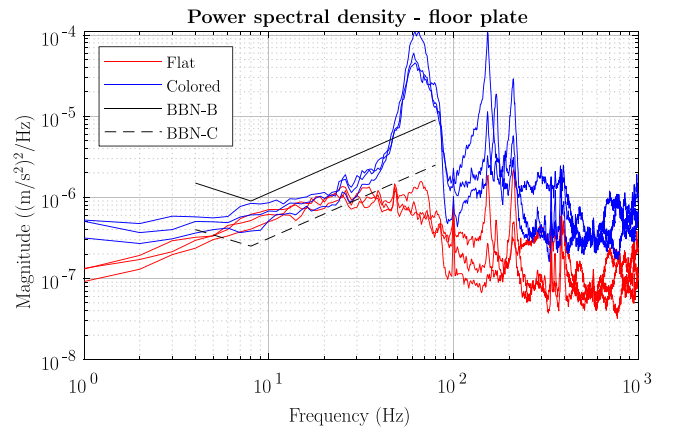


Fig. 2. Power spectral density (PSD) of the measured flat and colored floor vibration spectra induced by the piezo shakers, plus a comparison to the BBN-B/C specification curves from literature representing industrial environments.

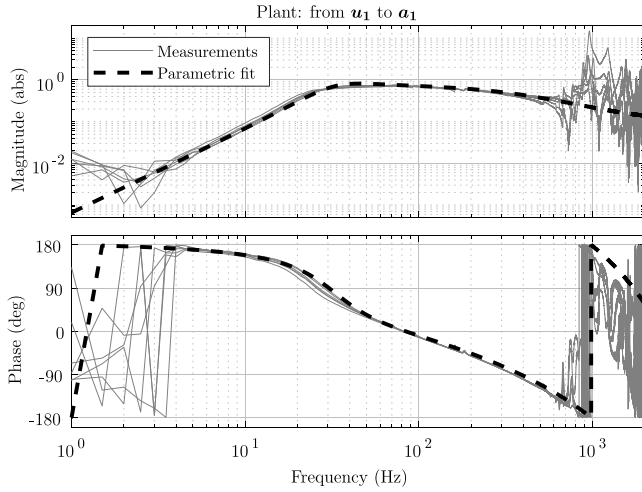


Fig. 3. Main diagonal elements of the measured frequency response matrix (FRM) of the system with feedback control, using u as inputs and a_1 as outputs; the parametric fit (dashed black) is obtained from discretization of the continuous transfer function $\hat{P}_2$ in (2) using the Tustin method and adding one sample delay.

system does not hit its limits. The design of the setup in Fig. 1 is such that its 6×6 transfer matrix from actuator voltages u to platform accelerations a_1 is diagonally dominant. Thereby a diagonal feedback controller is applied. Skyhook damping is added by proportional feedback of the platform velocities, which are obtained by integrating the inputs (accelerations) using weak integrators with a cut-off frequency of $\omega_i = 1$ rad/s. Furthermore a notch filter is used to avoid excitation of a high-frequency resonance at $\omega_n = 2\pi \cdot 962$ rad/s. This feedback control is implemented as

$$C_{FB}(s) = \frac{\omega_i}{s + \omega_i} k_v \frac{1 + \frac{1}{100}\omega_n + \omega_n^2}{1 + \omega_n + \omega_n^2} \mathbf{I}_6. \quad (1)$$

The feedback gain is tuned to $k_v = 225$ Ns/m, resulting in a relative damping of the suspension modes of about 0.7 for all six suspension modes. This tuning is based on the measured frequency response matrix (FRM) of the system with feedback discussed hereafter.

The diagonal elements of the measured FRM of the system with skyhook damping is shown in Fig. 3. The six lines are rather similar as a result of the axi-symmetric system design. A parametric fit of the FRM is obtained by manually fitting a physics-based model of the electromechanical dynamics to these diagonal elements, discretization of this model using the Tustin method and adding one sample delay (sample time $T_s = 1/f_s = 2 \cdot 10^{-4}$ s). The physics-based model of the electromechanical system is

$$\hat{P}_2(s) = \frac{s^2}{ms^2 + ds + k} \frac{\omega_a}{s + \omega_a} \mathbf{I}_6, \quad (2)$$

with equivalent mass $m = 1.4$ kg, damping $d = 290$ Ns/m, stiffness $k = 60 \cdot 10^3$ N/m and actuator pole $\omega_a = 2\pi \cdot 300$ rad/s. Input voltages are scaled to force units at DC, to ease interpretability of the parameters. Note that the parametric fit resembles the magnitude of the measured FRM up to about 500 Hz, which is beyond frequency range of interest. The deviations beyond this frequency are mainly due to parasitic dynamics of the platform suspension, which result in a high resonance peak at 962 Hz. The parametric fit of the transfer function from actuator voltages u to platform accelerations a_1 is used in Section 4 for feedforward controller design.

3. Problem formulation

In this paper, multi-input multi-output feedforward control is considered for which the block diagram is shown in Fig. 4. The measured

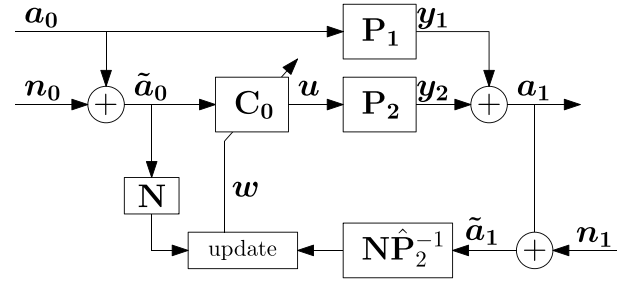


Fig. 4. Block diagram of a vibration isolation system with self-tuning disturbance feedforward control. In this figure a_0 is the floor acceleration with additive noise n_0 and a_1 is the payload acceleration with additive noise n_1 . P_1 is the primary path with output y_1 , P_2 is the secondary path with output y_2 , $\hat{P}_2$ the estimate of the secondary path, N is the noise shaping filter and C_0 is the self-tuning feedforward controller with output u and parameter vector w .

floor acceleration a_0 is considered as input disturbance while the measured payload acceleration $a_1 = y_1 + y_2$ is considered as the signal to be minimized in some norm-based sense. The latter is formed by two signals: y_1 is caused by a_0 via the primary path P_1 , and y_2 represents the impact of the control signal u transferred via the secondary path P_2 . The feedback controller (1) is considered as an integral part of P_1 and P_2 . The self-tuning feedforward controller C_0 provides disturbance feedforward control based on the measured signal $\tilde{a}_0 = a_0 + n_0$, denoting the floor plate acceleration with additive measurement noise represented by $n_0(t) \in \mathbb{R}^3$. Similarly, sensor noise $n_1(t) \in \mathbb{R}^6$ is added to the measured signal $\tilde{a}_1 = a_1 + n_1$. For fixed parameters in C_0 , a_1 can be written in the Laplace domain as

$$A_1(s) = \mathcal{T}(s)A_0(s) + S_0(s)N_0(s), \quad (3)$$

with transmissibility matrix $\mathcal{T}$ and noise sensitivity matrix S_0 which are defined as

$$\mathcal{T}(s) = P_1(s) + P_2(s)C_0(s), \quad (4)$$

$$S_0(s) = P_2(s)C_0(s). \quad (5)$$

Here, $\mathcal{T}$ is a measure for vibration isolation performance, whereas S_0 is a measure for noise amplification.

The control problem can now be formulated as follows. Given disturbance A_0 and noise N_0 , find a feedforward controller C_0 such that the power of A_1 is minimized in the frequency range of interest (1–300 Hz). In the absence of N_0 , perfect cancellation is obtained if $\mathcal{T} = 0$ which requires

$$C_{0,opt}(s) = -P_2^{-1}(s)P_1(s). \quad (6)$$

This controller follows from substitution of (6) in (4). However, a discrete-time implementation of (6) could result in a non-causal and unstable feedforward controller due to time delays and non-minimum phase zeros in P_2 . In practice, the aim is therefore to obtain the best causal and stable approximation of (6), see, e.g., [4,8]. For the hard-mount system considered in this paper, such an approximation is derived in [16] by means of a feedforward controller using spring-damper compensation:

$$C_0(s) = -\frac{1}{s^2} (Ds + K), \quad (7)$$

with damping matrix $D \in \mathbb{R}^{6 \times 3}$ and stiffness matrix $K \in \mathbb{R}^{6 \times 3}$. The expression in (7) is extended in this paper by including the effect of the actuator pole, caused by the inductance of the voltage-controlled VCMs. This pole was previously included in the parametric fit of the secondary path in (2). Including the inverse of the actuator dynamics in the feedforward controller gives

$$C_0(s) = -\frac{1}{s^2} (As + I_6) (Ds + K), \quad (8)$$

with $\mathbf{A} \in \mathbb{R}^{6 \times 6}$ a diagonal matrix with the inverse of the actuator poles on its diagonals. Expanding the result gives

$$\mathbf{C}_0(s) = -\frac{1}{s^2} \left(\underbrace{\mathbf{A}\mathbf{D}}_{\tilde{\mathbf{M}}} s^2 + \underbrace{(\mathbf{D} + \mathbf{A}\mathbf{K})}_{\tilde{\mathbf{D}}} s + \underbrace{\mathbf{K}}_{\tilde{\mathbf{K}}} \right) \quad (9)$$

with $\tilde{\mathbf{M}} \in \mathbb{R}^{6 \times 3}$, $\tilde{\mathbf{D}} \in \mathbb{R}^{6 \times 3}$ and $\tilde{\mathbf{K}} \in \mathbb{R}^{6 \times 3}$ the controller parameters. Any varying or unknown gains of the actuators or sensors can be absorbed in these controller parameters as well.

The controller in (9) is not feasible in practice because it induces drift and actuator saturation as a result of using pure integrators. To cope with this problem, it is proposed in [16] to replace the pure integrators by n th-order weak integrators. That is, the integrating actions are cut off below a certain frequency α . By doing so, the controller is given by

$$\mathbf{C}_{0,n}(s) = -(\tilde{\mathbf{M}} + H(s)\tilde{\mathbf{D}} + H(s)^2\tilde{\mathbf{K}}), \quad (10)$$

with

$$H(s) = \frac{1 - L_{(\alpha,n)}(s)}{s}, \quad (11)$$

that describes an n th-order weak integrator with relative degree one, and

$$L_{(\alpha,n)}(s) = \left(\frac{\alpha}{s + \alpha} \right)^n, \quad (12)$$

representing an n th-order low-pass filter with cut-off frequency α . As a result, an n th-order roll-off is obtained in $\mathcal{T}$ for frequencies beyond α . Here, the cut-off frequency $\alpha = 2\pi \cdot 2$ rad/s (2 Hz) and $n = 3$ are taken.

In this paper, the controller parameters in terms of $\tilde{\mathbf{M}}$, $\tilde{\mathbf{D}}$ and $\tilde{\mathbf{K}}$ in (10) are obtained using self-tuning for three reasons. First, self-tuning prevents the need of doing extensive system identification experiments to get sufficiently accurate estimations of the controller parameters. Second, self-tuning automatically accounts for the contribution of sensor noise $\mathbf{N}_0$ in the minimization of $\mathbf{a}_1$, which was neglected in the derivation of (10). Third, self-tuning enables online controller adaptations to track time-varying system parameters or changes in the input vibration spectrum. Self-tuning is explained in detail in the next section.

4. Self-tuning controller

This section presents the self-tuning FeRLS algorithm in discrete-time, as well as the FeLMS algorithm which is used for comparison of performance. The parameters will be updated at every time sample $t_k = kT_s$, with index $k \in \mathbb{N}$ and sampling time T_s . Both FeLMS and FeRLS use the same controller structure adopted from [16], where it was introduced for the FeLMS algorithm. This structure is summarized first. Afterwards, the self-tuning algorithms are presented.

4.1. Controller structure

The controller structure is derived from the continuous-time form in (10). Defining q representing the backward shift operator, a self-tuned feedforward control signal is defined as

$$\mathbf{u}_{FF}(k) = \underbrace{\left[-\tilde{\mathbf{M}}(k) \quad -\frac{1}{\beta}\tilde{\mathbf{D}}(k) \quad -\frac{1}{\beta^2}\tilde{\mathbf{K}}(k) \right]}_{\mathbf{W}(k)} \underbrace{\begin{bmatrix} \tilde{\mathbf{a}}_0(k) \\ \beta H(q)\tilde{\mathbf{a}}_0(k) \\ \beta^2 H(q)^2\tilde{\mathbf{a}}_0(k) \end{bmatrix}}_{\tilde{\Psi}(k)}. \quad (13)$$

The matrix $\mathbf{W}(k) \in \mathbb{R}^{6 \times 9}$ consists of the self-tuning controller gain matrices $\tilde{\mathbf{M}}(k)$, $\tilde{\mathbf{D}}(k)$ and $\tilde{\mathbf{K}}(k)$, and $\tilde{\Psi}(k) \in \mathbb{R}^9$ represents the regression vector. The constant $\beta \in \mathbb{R}$ is chosen such that the power of the product $\beta H(q)$ is normalized, which is shown in [16] to be optimal for FeLMS convergence. For a practical implementation, (13) is rewritten such that

the parameters from $\mathbf{W}(k)$ are stored in a single column vector $\mathbf{w}(k)$, or

$$\mathbf{u}_{FF}(k) = H(q) \cdot \underbrace{\begin{bmatrix} \tilde{\Psi}^T(k) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \tilde{\Psi}^T(k) \end{bmatrix}}_{\tilde{\Psi}(k)} \underbrace{\begin{bmatrix} (\mathbf{W}_{(1,:)}(k))^T \\ \vdots \\ (\mathbf{W}_{(6,:)}(k))^T \end{bmatrix}}_{\mathbf{w}(k)}, \quad (14)$$

with regression matrix $\tilde{\Psi}(k) \in \mathbb{R}^{6 \times 54}$, parameter vector $\mathbf{w}(k) \in \mathbb{R}^{54}$ containing the self-tuning controller parameters using either FeLMS or FeRLS, and $(\mathbf{W}_{(i,:)}(k))^T$ denoting the transposed i th row of $\mathbf{W}(k)$.

4.2. Filtered-error LMS

Stochastic gradient algorithms based on least mean squares (LMS) are derived from a steepest descent method satisfying the update law [8,10]

$$\mathbf{w}(k+1) = \mathbf{w}(k) - \mu(k) \left(\frac{\partial J}{\partial \mathbf{w}} \right)^T, \quad (15)$$

with possibly time-varying step-size $\mu(k)$, and cost function $J = \frac{1}{2} \mathbb{E}[e^T(k)e(k)]$ representing a mean squared filtered error. For the specific case of FeLMS with residual noise shaping, the filtered error is given by

$$e(k) = \mathbf{N}(q)\hat{\mathbf{P}}_2^{-1}(q)\tilde{\mathbf{a}}_1(k). \quad (16)$$

To obtain the filtered error in (16), the measured accelerations are filtered by the inverse of the secondary path denoted by $\hat{\mathbf{P}}_2^{-1}(q)$, in line with the use of filtered-error LMS. The additional filter $\mathbf{N}$ is used for residual noise shaping and specifies the frequency range of interest for vibration isolation. The model inverse of the secondary path $\hat{\mathbf{P}}_2^{-1}(q)$ is obtained by subsequent Tustin discretization and inversion of the parametric fit in (2). Since this parametric fit has more poles than zeros, inversion would lead to a non-proper system. Therefore, the discretization and inversion are combined with the multiplication with $\mathbf{N}$ which will have sufficient relative degree to make the product $\mathbf{N}\hat{\mathbf{P}}_2^{-1}$ proper.

The role of $\mathbf{N}$ is to remove signal contents at very low and high frequencies where model errors are expected to occur and which are outside the frequency range of interest [16]. Therefore, $\mathbf{N} = N(q)\mathbf{I}$ is designed as a diagonal filter containing a 3rd order high-pass filter at 6 Hz and a 3rd order low-pass filter at 300 Hz. This filter differs from [16] by a higher low-pass filter frequency, which is allowed by a reduction of the model error as a result of including the actuator pole in the dynamics (see (2) and (8)). Furthermore notch filters are added to $N(q)$ to suppress the strong response at the 962 Hz resonance and electrical power grid induced disturbances at 50, 100 and 200 Hz. Furthermore, the filter $\frac{s^2}{ms^2 + ds + k}$ is added to compensate the change in magnitude characteristics induced by $\hat{\mathbf{P}}_2^{-1}$.

LMS algorithms use an instantaneous approximation of the gradient in (15) by minimizing the instantaneous squared error, i.e. $J(k) = \frac{1}{2} e^T(k)e(k)$. Given (16), it is derived in [16] that, under the assumptions of slow adaptation and perfect secondary path estimation $\hat{\mathbf{P}}_2 = \mathbf{P}_2$, the gradient is approximated as

$$\left(\frac{\partial J(k)}{\partial \mathbf{w}} \right)^T = \tilde{\Psi}_N(k)e(k), \quad (17)$$

with filtered regressors $\tilde{\Psi}_N(k) = \mathbf{N}(q)\tilde{\Psi}(k)$. Next, define a normalized adaptation rate,

$$\mu(k) = \frac{\bar{\mu}}{\epsilon + \text{Tr}(\tilde{\Psi}_N^T(k)\tilde{\Psi}_N(k))}, \quad (18)$$

based on the trace of the instantaneously estimated covariance matrix of $\tilde{\Psi}_N$ and a small regularization constant $\epsilon \in \mathbb{R}$. Substitution of (17) and (18) in (15) yields the update law for FeLMS with residual noise shaping:

$$\mathbf{w}(k+1) = \mathbf{w}(k) - \frac{\bar{\mu}}{\epsilon + \text{Tr}(\tilde{\Psi}_N^T(k)\tilde{\Psi}_N(k))} \tilde{\Psi}_N^T(k)e(k). \quad (19)$$

4.3. Filtered-error RLS

Recursive least squares (RLS) algorithms provide the *exact* solution to a well-defined estimation problem with an exponentially weighted least-squares cost function $J(k) = \frac{1}{2} \sum_{j=1}^k \lambda^{k-j} e^T(j) e(j)$. Here, the forgetting factor $0 < \lambda \leq 1$ introduces exponential weighting to assign less weight to past samples such that it can track changes.

Alternative approaches, derived by [8,12], consider RLS as a stochastic gradient algorithm based on Newton's method:

$$\mathbf{w}(k+1) = \mathbf{w}(k) - \mu(k) \mathbf{A}^{-1}(k) \left(\frac{\partial J(k)}{\partial \mathbf{w}} \right)^T. \quad (20)$$

Similar to the steepest-descent method in (15), some step-size $\mu(k)$ and cost function $J = \frac{1}{2} \mathbb{E}[e^T(k) e(k)]$ are used. In contrast to (15), the method in (20) contains an additional scaling with the inverse of $\mathbf{A}$, representing the Hessian matrix of J with respect to $\mathbf{w}$. This is useful because it compensates for eigenvalue spread in the auto-correlation matrix of the regressors, which is known to be a main cause for non-uniform convergence speeds of the parameters in LMS algorithms [14].

To arrive at the FeRLS algorithm from (20), for the Hessian matrix $\mathbf{A}(k) = \mathbb{E}[\tilde{\Psi}_N^T(k) \tilde{\Psi}_N(k)]$, an exponentially weighted sample average is taken [12]

$$\mathbf{A}(k) = \alpha(k) \left[\lambda^{k+1} \epsilon \mathbf{I} + \sum_{j=0}^k \lambda^{k-j} \tilde{\Psi}_N^T(j) \tilde{\Psi}_N(j) \right], \quad (21)$$

where the gain $\alpha(k)$ is defined as

$$\alpha(k) = \begin{cases} \frac{1-\lambda}{1-\lambda^{k+1}} & \text{for } 0 < \lambda < 1 \\ \frac{1}{1+k} & \text{for } \lambda = 1 \end{cases}. \quad (22)$$

The gain $\alpha(k)$ is introduced to get the proper exponentially weighted sample average. The gain for $\lambda = 1$ gives the normal sample average [12], while the gain for $0 < \lambda < 1$ is derived in Appendix B. Eq. (21) provides an accurate estimation for the co-variance matrix, where the regularization constant ϵ will automatically vanish for increasing k . Next, to obtain the FeRLS algorithm, the adaptation gain is taken as

$$\mu(k) = \alpha(k), \quad (23)$$

where $\alpha(k)$ is defined in (22). The adaptation gain is 1 for $k = 0$ and decreases for increasing k . The adaptation gain decreases to the limit $1 - \lambda$ as $k \rightarrow \infty$ and thus a vanishing adaptation gain is obtained for $\lambda = 1$, while the adaptation gain remains nonzero for smaller λ .

Substitution of (17), (23) and (22) in (20) (for $0 < \lambda < 1$) yields

$$\mathbf{w}(k+1) = \mathbf{w}(k) - \frac{1-\lambda}{1-\lambda^{k+1}} \mathbf{A}^{-1}(k) \tilde{\Psi}_N^T(k) e(k), \quad (24)$$

and further substitution of (21) in (24) gives

$$\mathbf{w}(k+1) = \mathbf{w}(k) - \tilde{\mathbf{P}}(k) \tilde{\Psi}_N^T(k) e(k), \quad (25)$$

with adaptation gain matrix

$$\begin{aligned} \tilde{\mathbf{P}}(k) &= \mu(k) \mathbf{A}^{-1}(k) \\ &= \left[\lambda^{k+1} \epsilon \mathbf{I} + \sum_{j=0}^k \lambda^{k-j} \tilde{\Psi}_N^T(j) \tilde{\Psi}_N(j) \right]^{-1}. \end{aligned} \quad (26)$$

Note that the inclusion of the exponential weighting makes sure $\tilde{\mathbf{P}}(k)$ does not go to zero. For a practical implementation, direct use of (25) is not desired as it implies the inversion of a 54×54 matrix to obtain $\tilde{\mathbf{P}}(k)$ at every time step. This computationally heavy operation is avoided by taking two steps to turn (24) into an efficient FeRLS algorithm. These steps are discussed hereafter and elaborated in Appendix A.

First, due to the block diagonal structure of $\tilde{\Psi}_N(j)$ (see Eq. (14)), the multi-input multi-output adaptation gain matrix has a repeating block-diagonal structure as well:

$$\tilde{\mathbf{P}}(k) = \begin{bmatrix} \mathbf{P}(k) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \mathbf{P}(k) \end{bmatrix}, \quad (27)$$

where $\mathbf{P}(k) \in \mathbb{R}^{9 \times 9}$ is defined as

$$\mathbf{P}(k) = \left[\lambda^{k+1} \epsilon \mathbf{I} + \sum_{j=0}^k \lambda^{k-j} \tilde{\Psi}_N(j) \tilde{\Psi}_N^T(j) \right]^{-1}. \quad (28)$$

Second, the matrix inversion lemma is used to turn (25) into the recursive update

$$\mathbf{P}(k) = \lambda^{-1} [\mathbf{P}(k-1) - \frac{\mathbf{P}(k-1) \tilde{\Psi}_N(k) \tilde{\Psi}_N^T(k) \mathbf{P}(k-1)}{\lambda + \tilde{\Psi}_N^T(k) \mathbf{P}(k-1) \tilde{\Psi}_N(k)}]^{-1}, \quad (29)$$

with filtered regressors $\tilde{\Psi}_N(k) = N(q) \tilde{\Psi}(k)$. The algorithm is initialized by $\mathbf{P}(-1) = \epsilon^{-1} \mathbf{I}$, where $\epsilon \in \mathbb{R}$ should be chosen as a small number. Note that no matrix inverse needs to be calculated at all. The ability to use a scalar division is the result of the block-diagonal parametrization of the feedforward controller in (14).

The block-diagonal structure and repetitions in $\tilde{\mathbf{P}}(k)$ and $\tilde{\Psi}_N(k)$ are exploited further to improve computational efficiency by reformulating (25) to

$$\mathbf{w}(k+1) = \mathbf{w}(k) - \begin{bmatrix} \mathbf{P}(k) \tilde{\Psi}(k) & \dots & \mathbf{P}(k) \tilde{\Psi}(k)^T \end{bmatrix} e(k), \quad (30)$$

where $\mathbf{P}(k)$ and $\mathbf{P}(k) \tilde{\Psi}(k)$ only need to be computed once. This efficiency exploits the fact that each input is affected by the same regressor vector as shown in (14).

4.4. FeRLS versus FeLMS

Two key differences can be distinguished by comparing the FeRLS update law (24) to (19) for FeLMS. First, it is observed that FeLMS has a fixed adaptation rate $\tilde{\mu}$ such that the algorithm keeps on updating. As a consequence, FeLMS will remain a non-zero variance on the estimated parameters in steady-state. In contrast, FeRLS has a decreasing adaptation gain resulting in smaller parameter variance. The variance on the estimated parameters completely disappears for $\lambda = 1$, because the adaptation gain vanishes completely. For $0 < \lambda < 1$ tracking ability is preserved at cost of a (small) remaining variance of the estimated parameters.

Second, FeLMS in fact satisfies Newton's method when one defines

$$\mathbf{A}^{-1}(k) = \frac{54}{\epsilon + \text{Tr}(\tilde{\Psi}_N^T(k) \tilde{\Psi}_N(k))} \mathbf{I}_{54} \quad (31)$$

as an instantaneous estimated inverse of the Hessian matrix $\mathbf{A} = \mathbb{E}[\tilde{\Psi}_N^T(k) \tilde{\Psi}_N(k)]$. However, this instantaneous estimate used in FeLMS will be less accurate compared to the Hessian used in FeRLS for two reasons. First, it assumes that the regressor signals in $\tilde{\Psi}_N$ are orthogonal (cross-terms are zero) and all have the same power ($\mathbf{A}$ in (31) has 54 identical diagonal elements). These properties are unlikely for stochastic signals and a frequency-dependent residual noise shaping filter. Second, it approximates the expected value of the power by a regularized instantaneous estimate, i.e., $\mathbb{E}[\tilde{\Psi}_N^T(k) \tilde{\Psi}_N(k)] \approx (\epsilon + \text{Tr}(\tilde{\Psi}_N^T(k) \tilde{\Psi}_N(k))) \mathbf{I}$, which is not correct for stochastic signals satisfying a probability density function. Therefore, FeLMS will in general not have the fast and uniform convergence properties such as obtained with FeRLS.

5. Experimental validation

This section presents an experimental validation to demonstrate the feasibility of the proposed FeRLS algorithm and the performance improvements obtained with FeRLS compared to FeLMS. First, the transient performance of the vibration isolation system is studied. Second, steady-state vibration isolation performance after convergence is evaluated using plots of the transmissibility matrix and the power spectral densities.

Table 1

Data obtained from the measured convergence of the first nine parameters using FeLMS or FeRLS. The time constant τ (where the signal remains within 63% of its final value) and the final value v_{final} are listed.

Quantity	Algorithm	Parameter								
		1	2	3	4	5	6	7	8	9
τ (s)	FeLMS	67.4	47.5	37.4	13.2	6.6	9.6	7.3	16.4	24.7
	FeRLS	39.0	26.2	1.0	7.8	3.3	6.1	1.8	3.1	4.3
v_{final}	FeLMS	-0.0320	0.0659	0.2543	0.1198	0.3808	-0.3944	0.5162	1.5028	-0.7239
	FeRLS	-0.0180	0.0884	0.2151	0.1404	0.3383	-0.3720	0.5321	1.5175	-0.7648

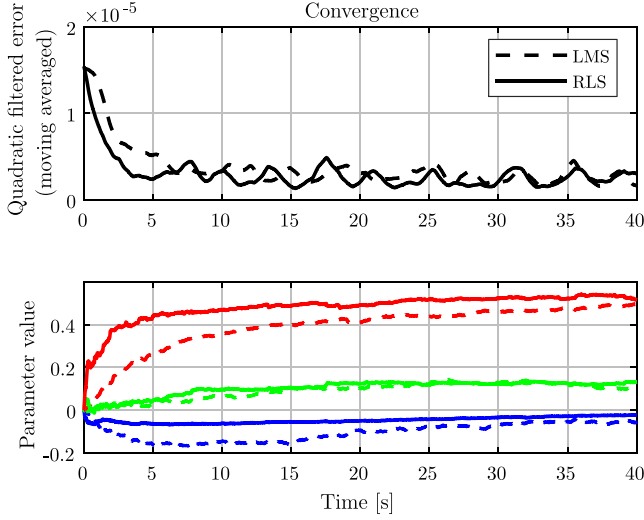


Fig. 5. Measured convergence of the quadratic filtered error and some characteristic parameters (w_1 in blue, w_4 in green and w_7 in red) using either FeLMS (dashed) or FeRLS (solid). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.1. Transient performance

Transient performance is analyzed by observing both the initial convergence behavior and the ability to respond to changing disturbance frequency spectra.

5.1.1. Initial convergence

Self-learning is initialized as follows. For FeLMS, numerical values $\bar{\mu} = 10^{-4}$ and $\epsilon = 10^{-5}$ are used. For FeRLS, the initial covariance matrix $\mathbf{P}_0$ is set to a diagonal matrix with values equal to 10 times the steady-state values to speed up initial convergence, and the forgetting factor is set to $\lambda = \exp\left(-\frac{1}{10}T_s\right)$, providing adaptation capabilities with a time-constant of 10 s. This choice of the forgetting factor eventually results in similar steady-state parameter variance for FeRLS and FeLMS as will be shown later. Here, the flat floor vibration spectrum from Fig. 2 is used.

Convergence of the quadratic filtered error and some characteristic parameters is shown in Fig. 5 for both FeLMS and FeRLS. Table 1 shows the results from convergence in terms of a time constant and the final value. The table shows the first nine parameters, corresponding to the first control signal in u . Due to the system's symmetry, the other parameters show similar trends and have not been included for space limitations.

It is observed that both algorithms converge to about the same values. Minor differences appear because both algorithms minimize a slightly different cost function. Both algorithms show significant differences in convergence speed. FeLMS suffers from eigenvalue spread introduced by the instantaneous estimate of the Hessian matrix, see (31), because the floor vibration spectrum is not perfectly flat. FeRLS suffers from the fact that the initial estimate for the $\mathbf{P}_0$ matrix is not

perfect. Nevertheless, the time constants for FeRLS are much faster (1.7 till 37 times, see Table 1), demonstrating the superiority of FeRLS. The steady-state variance of the parameters is similar for both algorithms, which can be seen from Fig. 5. One could decrease this variance in FeLMS by setting a lower convergence rate $\bar{\mu}$, but this will further decrease the convergence rate. In FeRLS the variance can be decreased by taking λ closer to 1; taking $\lambda = 1$ even results in convergence in mean square, due to the vanishing adaptation rate. However, this is also at cost of convergence speed and the ability to track changes in the disturbances.

5.1.2. Disturbance tracking

Next, a comparison between FeLMS and FeRLS is made in view of tracking a change in the floor vibration spectrum. To illustrate this, the floor spectrum is instantaneously changed from a flat to a colored spectrum (as defined in Fig. 2) and the resulting response in parameter convergence has been measured, see Fig. 6(a). It is observed that the parameters converge faster to the new situation when FeRLS is used. In Fig. 6(b), a similar measurement is shown but then for a switch from a colored spectrum back to a flat spectrum. In this case, the benefit of FeRLS is no longer there. This is because FeRLS first has to re-adapt the $\mathbf{P}_0$ matrix to the flat spectrum whereas FeLMS directly benefits from the fact that the system more or less satisfies the strict assumptions for optimal FeLMS convergence speed (i.e., an approximately flat frequency spectrum, orthonormal basis functions, and proper choices for $\bar{\mu}$ and ϵ). Nevertheless, since FeRLS does not require these assumptions at all while still outperforming FeLMS in case of colored disturbances, FeRLS in general gives the best transient performance.

In these experiments, the computational loads were similar, i.e., 0.14 ms, although one might expect that FeRLS would need a higher computational load. This indicates that the update algorithm itself is not the most critical step in terms of computational load.

5.2. Steady-state performance

Steady-state performance will be evaluated in terms of transmissibility and (cumulative) power spectral density plots. For ease of presentation, both the floor plate and platform accelerations are transformed to a central Cartesian coordinate frame as defined in Fig. 1, and results are shown in the directions z, θ_x, θ_y .

5.2.1. Transmissibilities

Fig. 7 shows the measured transmissibility matrix for the directions z, θ_x , and θ_y . Results are shown for the passive, the feedback-controlled and the feedback- plus feedforward-controlled case using either FeLMS or FeRLS. The transmissibilities are measured using the flat spectrum with fixed parameters after convergence. It is observed that FeLMS and FeRLS result in similar reductions in magnitude after self-tuning. This is as expected, because it was previously observed that the parameters in essence converge to the same value (see Fig. 5 and Table 1). The main diagonals of the transmissibility matrix associate with reductions beyond 20 dB and up to 40 dB in all three directions with respect to the feedback-only controlled system for frequencies between 3 and 200 Hz. Beyond 200 Hz, performance is slightly deteriorated due to

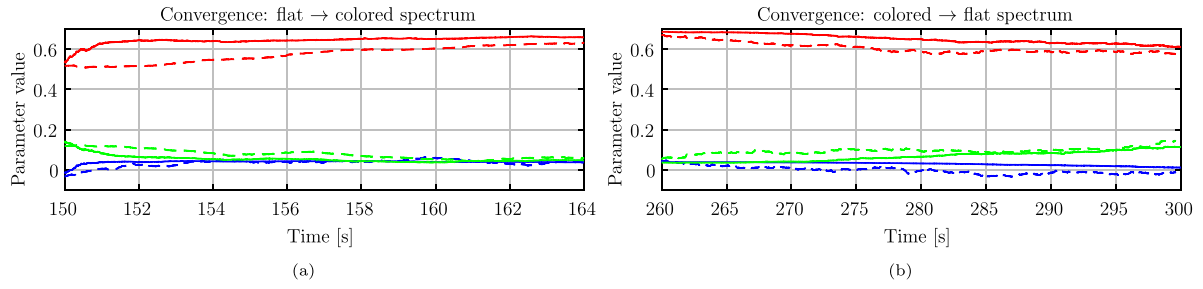


Fig. 6. (a) Response in parameter convergence from flat to colored floor spectrum, and (b) vice versa; the solid lines correspond to FeRLS, the dashed lines correspond to FeLMS.

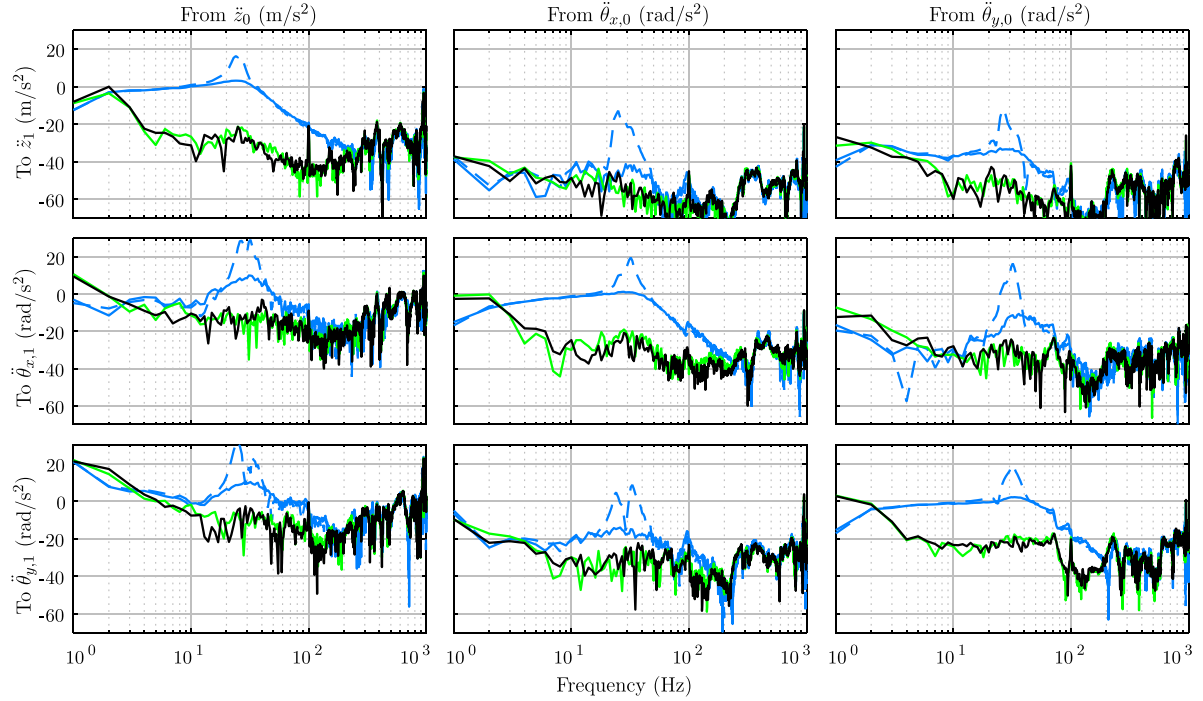


Fig. 7. Measured vibration isolation performance in terms of the transmissibility functions; performance is shown for the passive system (blue dashed), for the active system with feedback (blue), and for the active systems with feedback plus feedforward using either FeLMS (green) or FeRLS (black). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

a waterbed effect caused by the causal nature of disturbance feedforward control, i.e. the floor disturbance occurs prior to detection and compensation [27,28]. The off-diagonal terms in the transmissibility matrix have different lower bounds because not all inputs and outputs have the same units (meters vs. radians). For frequencies below 2 Hz, performance is slightly deteriorated due to application of the weak integrators [16]. A remarkable observation is that the low-frequency transmissibilities for the passive system and the active system with feedback are not 0 dB at low frequencies, probably due to the low excitation level and high sensor noise levels at these frequencies. For frequencies beyond 400 Hz, sensor noise and internal eigenmodes of the floor plate dominate the response. At 100 Hz some sharp peaks are visible, which were caused by disturbances from the electrical power grid.

5.2.2. Residual output power

Recall the main objective of this paper defined in Section 3, i.e., minimization of the platform accelerations $\mathbf{a}_1$ within the frequency range of interest (1–300 Hz). To evaluate this objective, Fig. 8(a) shows measured power spectral density (PSD) plots of $\mathbf{a}_1$ for the different control cases. It is observed that within the frequency range of 3–200 Hz, both FeLMS and FeRLS provide significant performance improvements

up to the theoretical limit given by the output noise covariance matrix.¹ Again, the deteriorated performance below 2 Hz is due to the higher-order weak integrators. Beyond 300 Hz, no performance improvements up to the theoretical limit are obtained. This can be explained by the fact that these frequencies are beyond the high cut-off frequency (300 Hz) of the band-pass filter in the noise shaping filter N. As such, the self-tuning algorithm does not optimize for such high frequencies, but allows here for performance deterioration which is inevitable due to the earlier mentioned waterbed effect.

Fig. 8(b) presents cumulative PSD plots, showing that the addition of feedforward control leads to significant performance improvements. Compared to the system using only feedback control, the cumulative power at 300 Hz has reduced with 94 % in the z -direction, with 90 % in the θ_x -direction, and with 93 % in the θ_y -direction in case of FeRLS-feedforward control. Similar reductions are found in case of FeLMS-feedforward control, which is expected because the parameters obtained by FeLMS and FeRLS in essence converge to the same global minimum.

¹ Under the assumption that the output noise is uncorrelated with $\mathbf{a}_0$, it is shown in [4] that the output noise covariance matrix can be calculated from spectral analysis techniques, and that this matrix indicates a lower bound for performance improvements possible with disturbance feedforward control

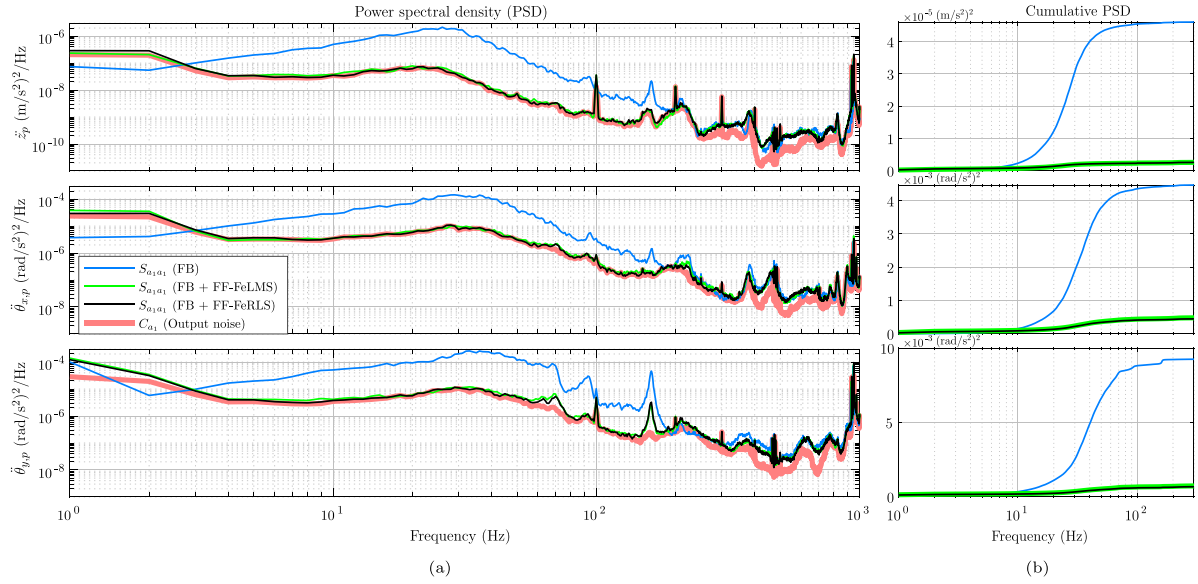


Fig. 8. (a) measured power spectral density (PSD) plots of the platform accelerations, and (b) measured cumulative PSD plots, for the active system with feedback (blue), and feedback plus FeLMS-feedforward (green) or FeRLS-feedforward (black). The PSD can be compared to the output noise, which is the theoretical minimum obtained by the used feedforward approach. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6. Conclusions

The FeRLS algorithm with residual noise shaping for self-tuning disturbance feedforward control is proposed for multi-axis vibration isolation. It is shown that the estimation of the input covariance makes orthonormalization of the basis functions or whitening of the input signal obsolete.

An efficient algorithm for FeRLS is derived by exploiting the sparsity of the covariance matrix. This allows online estimation of 54 parameters at a sample rate of 5000 Hz.

The FeRLS algorithm with residual noise shaping is implemented on a multi-axis hard-mount vibration isolation platform. The parameters estimated by FeRLS show a higher convergence speed compared to FeLMS. The forgetting factor of FeRLS is tuned to have similar parameter variance. The higher convergence speed allows faster deployment and tracking of changes in the parameters. After convergence similar parameters are found for FeLMS and FeRLS, giving similar vibration isolation performance.

In addition to previous work, the actuator dynamics are included in the dynamic description of the secondary path, which allows to approximate the dynamics accurately up to 500 Hz. This allows to increase the frequency band of the noise-shaping filter in the filtered-error approach up to 300 Hz, thereby increasing the frequency band for minimization of the (filtered) error. The transmissibility to floor vibrations is reduced from 20 dB up to 40 dB in the frequency range of interest compared to the feedback-only isolation.

The eventual vibration isolation of both FeRLS and FeLMS is limited at low frequencies by the weak integrators [16] and at high-frequency performance by the waterbed effect caused by the causal nature of disturbance feedforward control [28]. Future research will consider optimal balancing of all noise sources and limiting effects to achieve optimal overall performance.

RLS algorithms are known to have limited numerical stability [12]. In the experiments no numerical stability issues were observed. The limited number of recursion samples, the floating-point implementation and the use of a forgetting factor tend to attenuate numerical instability, but probably do not eliminate it. Various methods for mitigating numerical instabilities have been proposed, e.g. [12,29]. Numerical stability and mitigation of any instabilities are subject of further research.

CRediT authorship contribution statement

Wouter B.J. Hakvoort: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – original draft. **Michiel A. Beijen:** Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Matrix inversion lemma

This section shows how the inverse of the Hessian matrix can be recursively computed. Define

$$\bar{\mathbf{P}}(k) = \left[\lambda^{k+1} \mathbf{e} \mathbf{I} + \sum_{j=0}^k \lambda^{k-j} \tilde{\Psi}_N^T(j) \tilde{\Psi}_N(j) \right]^{-1} \quad (\text{A.1})$$

As an intermezzo, note that the summation can be rewritten to

$$\begin{aligned} \sum_{j=0}^k \lambda^{k-j} \tilde{\Psi}_N^T(j) \tilde{\Psi}_N(j) &= \sum_{j=0}^k \lambda^{k-j} \begin{bmatrix} \tilde{\Psi}_N(j) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \tilde{\Psi}_N(j) \end{bmatrix} \begin{bmatrix} \tilde{\Psi}_N^T(j) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \tilde{\Psi}_N^T(j) \end{bmatrix} \\ &= \begin{bmatrix} \sum_{j=0}^k \lambda^{k-j} \tilde{\Psi}_N(j) \tilde{\Psi}_N^T(j) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \sum_{j=0}^k \lambda^{k-j} \tilde{\Psi}_N(j) \tilde{\Psi}_N^T(j) \end{bmatrix} \end{aligned}$$

Next, define

$$\mathbf{P}^{-1}(k) = \lambda^{k+1} \mathbf{e} \mathbf{I} + \sum_{j=0}^k \lambda^{k-j} \tilde{\Psi}_N(j) \tilde{\Psi}_N^T(j), \quad (\text{A.2})$$

such that

$$\bar{\mathbf{P}}(k) = \begin{bmatrix} \mathbf{P}^{-1}(k) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \mathbf{P}^{-1}(k) \end{bmatrix}^{-1} \quad (\text{A.3})$$

$$= \begin{bmatrix} \mathbf{P}(k) & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \mathbf{P}(k) \end{bmatrix}. \quad (\text{A.4})$$

In other words, to construct $\bar{\mathbf{P}}(k)$ it is sufficient to calculate $\mathbf{P}(k)$, which can be written as

$$\begin{aligned} \mathbf{P}(k) &= \left[\lambda^{k+1} \mathbf{e} \mathbf{I} + \sum_{j=0}^k \lambda^{k-j} \tilde{\psi}_N(j) \tilde{\psi}_N^T(j) \right]^{-1} \\ &= \lambda^{-1} \left[\underbrace{\lambda^k \mathbf{e} \mathbf{I} + \sum_{j=0}^{k-1} \lambda^{k-1-j} \tilde{\psi}_N(j) \tilde{\psi}_N^T(j)}_{\mathbf{P}^{-1}(k-1)} \right. \\ &\quad \left. + \underbrace{\lambda^{-1} \tilde{\psi}_N(k)}_{\mathbf{B}(k)} \underbrace{\tilde{\psi}_N^T(k)}_{\mathbf{C}(k)} \right]^{-1}. \end{aligned} \quad (\text{A.5})$$

Next, consider the matrix inversion lemma [8]:

$$[\mathbf{A} + \mathbf{B}\mathbf{C}]^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B} [\mathbf{I} + \mathbf{C}\mathbf{A}^{-1}\mathbf{B}]^{-1} \mathbf{C}\mathbf{A}^{-1},$$

which assumes that $[\mathbf{I} + \mathbf{C}\mathbf{A}^{-1}\mathbf{B}]$ is non-singular. Application of this lemma to (A.5) yields

$$\begin{aligned} \mathbf{P}(k) &= \lambda^{-1} \left[\mathbf{P}(k-1) - \frac{\mathbf{P}(k-1) \lambda^{-1} \tilde{\psi}_N(k) \tilde{\psi}_N^T(k) \mathbf{P}(k-1)}{1 + \tilde{\psi}_N^T(k) \mathbf{P}(k-1) \lambda^{-1} \tilde{\psi}_N(k)} \right] \\ &= \lambda^{-1} \left[\mathbf{P}(k-1) - \frac{\mathbf{P}(k-1) \tilde{\psi}_N(k) \tilde{\psi}_N^T(k) \mathbf{P}(k-1)}{\lambda + \tilde{\psi}_N^T(k) \mathbf{P}(k-1) \tilde{\psi}_N(k)} \right] \end{aligned}$$

Providing an efficient algorithm to compute $\mathbf{P}(k)$ as it involves no matrix inversion.

Appendix B. Exponentially weighted sample average

In this section the scaling gain $\alpha(k)$ of the exponentially weighted sample average is derived. Consider the exponentially weighted average of some signal $c(k)$

$$\bar{c}(k) = \alpha(k) \sum_{j=0}^k (\lambda^{k-j} c(j)), \quad (\text{B.1})$$

where a proper scaling gain $\alpha(k)$ should result in $\bar{c}(k) = c$ for a constant $c(k) = c$. A constant c can be taken out of the summation and thus the following should hold

$$\alpha(k) \sum_{j=0}^k \lambda^{k-j} = 1. \quad (\text{B.2})$$

Some rewriting, gives

$$\alpha(k) \sum_{j=0}^k \lambda^{k-j} = \alpha(k) \sum_{j=0}^k \lambda^j = \alpha(k) \frac{1 - \lambda^{k+1}}{1 - \lambda} = 1, \quad (\text{B.3})$$

where the well-known solution for a geometric series is used in the second step. The previous equality is satisfied for

$$\alpha(k) = \frac{1 - \lambda}{1 - \lambda^{k+1}}, \quad (\text{B.4})$$

providing the proper scaling gain for the weighted sample average.

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